

BUILDING CHORDS FROM STACKED THIRDS

Every chord quality is determined by which thirds are stacked. Two intervals create a triad; a third interval creates a 7th chord.

CHORD FORMULAS AS STACKED INTERVALS



Major 3rd (M3) = 4 semitones



Minor 3rd (m3) = 3 semitones



Root note

Major

C (C-E-G)



Minor

Cm (C-E $\flat$ -G)

Diminished

C $\sharp$ (C-E $\flat$ -G $\sharp$)

Augmented

C $\sharp$ (C-E-G $\sharp$)

Dom7

C7 (C-E-G-B $\flat$)

Maj7

C Δ 7 (C-E-G-B)

Min7

Cm7 (C-E $\flat$ -G-B $\flat$)

DIM7 SYMMETRY

C $\sharp$ 7: 3 stacked minor 3rdsm3 (G $\flat$ - A = dim7)m3 (E $\flat$ - G $\flat$)m3 (C $\flat$ - E $\flat$)

C

All 4 notes are equal. Any one can be "the root" — divides the octave into 4 equal parts.

ONLY 3 UNIQUE DIM7 SOUNDS

C - E $\flat$ - G $\flat$ - A

= 4 inversions of the same sound

C $\sharp$ - E - G - B $\flat$

= 4 inversions of the same sound

D - F - A $\flat$ - B

= 4 inversions of the same sound

The other 9 "dim7 chords" you might name are just inversions of these 3 sounds.